

Upstream-downstream bridge 0.1-0.4 | Review r1

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This collection reviews the finite bridge between upstream 0.10 and downstream 0.12. Both endpoints remain closed. Bridge research ends at 0.8; 1.0 consolidates its publication.

Stage 0.1 defines typed interfaces. Stage 0.2 executes a conditional address/internal state and compares product reduction. Stage 0.3 replaces the reference phenotype allocation with matched SI rest and excitation energies. Stage 0.4 derives pressure and homogeneous expansion from that energy.

Re-executed case checks: 15, 95, 134 and 600, plus 36 cross-stage checks: 880 total implementation/mathematical checks, not experiments.

Revision r1 rejects nonfinite and negative exchange inputs, adds block trace controls, scopes provenance to bundled dependencies and propagates newly generated results through the pipeline.

Completion means execution under disclosed assumptions. Preparation hardware, species abundance, source-frame seconds, durable physical records and nonuniform covariant dynamics remain open.

1 | Typed interfaces and correlated state

Address shares 0.05/0.268/0.682 differ from inherited SI energy shares 0.0493/0.265/0.6857.
Response calibration is explicit. Resident Actual 0.318 differs in scope from the complete ledger 1.

For phi-conditioned addresses $n=2..1,000,000$, the exact block representation is:

$$\text{Sigma_phi} = \sum_n p(n|\text{phi}) |n\rangle\langle n| \times \text{rho_internal}(n) \times I_{128/128}$$

$$\text{rho_internal}(n) = (1-0.2 \chi(n)) |g\rangle\langle g| + 0.2 \chi(n) |e1\rangle\langle e1|$$

χ selects odd prime-3 valuation. Two frozen modes occupy the real 95-dimensional nucleon basis; the stationary carrier is 128-dimensional. D and return keep their original branch ledgers without nucleon preparation. General entanglement and address coherence preservation are not claimed.

Address-independent internal energy and charge-current expectations equal those of the internal marginal product. Address-weighted probes retain a covariance term that the product loses.

For $r(n)=1+0.25 \log(n)/\log(N)$, the baseline excitation-probe difference is 0.1447722277 MeV. It equals $a \cdot \text{gap} \cdot (\langle r \chi \rangle - \langle r \rangle \langle \chi \rangle)$. This diagnostic is not an additional cosmic energy term. Fixed charge operators were tested at $Q^2=0, 0.01, 0.1 \text{ GeV}^2$.

2 | SI replacement and signed exchange

The exact conversion is $1 \text{ MeV} = 1.602176634 \times 10^{-13} \text{ J}$. Particle count is not an address probability. An external reference matching sets an expected occupancy in $V_0 = 1 \text{ m}^3$:

$$N_{\text{bar}} = E_{\text{phi_ref}} / (E_{\text{rest_particle}} + E_{\text{exc_baseline_particle}})$$

$$E_{\text{system}} = N_{\text{bar}} * E_{\text{rest_particle}} + N_{\text{bar}} * E_{\text{exc_case_particle}} + E_{\text{D}} + E_{\text{R}}$$

This decomposition replaces E_{phi} . Adding it to the original total again duplicates E_{phi} . Pure-proton and pure-neutron illustrations under two calibrations do not predict actual composition.

The inherited proton example has $N_{\text{bar}} = 0.2425893464$: an expected count, not an integer event count. Its rest and reference excitation totals are $3.6467913480 \times 10^{-11} \text{ J}$ and $1.3399990150 \times 10^{-12} \text{ J}$.

$\Delta E = N_{\text{bar}} * (E_{\text{exc_case}} - E_{\text{exc_baseline}})$. System and environment receive opposite entries. A reservoir initially equal to $0.2 * E_{\text{phi_ref}}$ is a disclosed finite diagnostic input. The low-amplitude conditional preparation gives $+8.9385339569 \times 10^{-15} \text{ J}$ to the system.

An external reservoir supplies boundary work. An included reservoir cancels internal exchange in the full energy ledger. The inherited admission conversion work is a separate event account; it is not added without checking event identity.

3 | Same-energy pressure and homogeneous response

After preparation, comoving population and internal state are fixed. Phenotype and D total energies are constant; R has constant density. An included reservoir additionally assumes pressure zero.

$$V=V_0*a^3; E(V)=M+L*V; P=-dE/dV=-L; u=M/V+L$$

$$H^2/H_0^2=u/u_{crit}; q=(u+3P)/(2u); a*du/da+3*(u+P)=0$$

At $a=1$ with an external reservoir, inherited $q=-0.52855$ and alternate $q=-0.523$ both reproduce $H/H_0=1$. At $a=.5, 1, 2$, q uses evolving energy shares.

Including the finite reservoir changes the reference load without refitting: inherited $q=-0.5185075159$, $H/H_0=1.0049179071$; alternate $q=-0.5128712871$, $H/H_0=1.0049875621$. These are conditional load results, not new observational calibrations.

The execution supplies 96 volume rows and 32 homogeneous integrations with 17 sample times in $\tau=H_0*t$. Maximum scale-factor error against the analytic dust/background solution is $3.297429e-11$. This time coordinate does not assign physical seconds to source frames.

Same-energy pressure, continuity and homogeneous acceleration agree. Real reservoir pressure, $Q(t)$ during preparation, microscopic volume coupling and nonuniform dynamics remain open. Local rotation and lensing were not recalculated.

4 | Reproduction and closure ledger

At package root run `OPENBLAS_NUM_THREADS=1 python reproduce_all.py`. Numerical dependencies are Python 3, numpy, scipy and mpmath. PDF rebuilding additionally needs reportlab and NanumGothic.

The runner regenerates 0.2 → 0.3 → 0.4 inputs sequentially and checks 36 cross-stage contracts. Original ZIPs remain in historical/. Current executable evidence is in studies/ and review_checks.json.

The implemented bridge covers a conditional phi state, external reference-matched expected populations, finite signed exchange and homogeneous post-preparation evolution. Simultaneous source-branch population transport and particle creation are not executed.

Remaining stages: 0.5 clocks and SI phase; 0.6 measurement, records and repeated outcomes; 0.7 finite nonuniform coupling; 0.8 complete replay and review. Scalar agreement does not close an unimplemented connection.

Upstream integrated source: DOI 10.5281/zenodo.23119041. Downstream integrated source and 0.10-0.12 lineage: DOI 10.5281/zenodo.23113101. The coauthored Physics preparation manuscripts are cited by those source collections.

Repository: <https://github.com/Wonsik-Choi-janefather/wrra-m-0.1>. Source/result hashes and a complete SHA-256 manifest accompany the release.